

Learning objectives

In this lesson you will learn to:

- convert between signed denary numbers and two's complement binary numbers
- determine the range of values that can be represented in two's complement by a binary number of a given length.

For more information about this topic and additional student activities see Topic 2 of the student book.

Two's complement

We discussed two's complement in a previous lesson.

What do you remember?

Two's complement is the most common way of representing signed binary integers.

There is only one way of representing 0.

This is better than sign magnitude representation. Can you remember why?

Recap

In two's complement, the **most significant bit (MSB)** is worth -128.

-128	64	32	16	8	4	2	1	

What is the pattern for the highest value that can be represented in 8 bits?

What is the pattern for the lowest value that can be represented in 8-bits?

Maximum and minimum values

-128	64	32	16	8	4	2	1	
0	1	1	1	1	1	1	1	

-128	64	32	16	8	4	2	1	
1	0	0	0	0	0	0	0	

What is two's complement and why is it needed?

Two's complement is actually a clever mathematical solution.

As we saw before, the human-friendly minus sign cannot be displayed in binary code.

Sign-magnitude representation has two values for 0. That's not particularly useful!

Two's complement is much more useful as there is only one value for 0 and mathematical operations actually work.

Adding with two's complement

Adding with binary patterns in two's complement is the same as adding any other binary pattern, including those that represent unsigned numbers.

The four rules for addition are the same

0	0	0	0	0	1	1	1			=	+7
1	1	1	1	1	0	1	1	+		=	-5
0	0	0	0	0	0	1	0			=	+2

Wrap up: so far you have learned how to...

- ✓ Convert between signed denary numbers and two's complement binary numbers
 - Remember that the MSB is its negative equivalent.
- ✓ Determine the range of values that can be represented in two's complement by a binary number of a given length.
 - Maximum value is 0 in MSB and 1s in every other bit.
 - Minimum value is 1 in MSB and 0s in every other bit.